

Reproducibility and open-source research workflow

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graph TD; A[Reproducibility and open-source research workflow] --> B["Synthetic data generation<br/>(controlled uncertainty, Python modules)"]; B --> C["Deterministic logic execution<br/>(TRI-X core inference)"]; C --> D["XAI analysis<br/>(SHAP / NMF / counterfactual / trace)"]; D --> E["Safety-oriented evaluation<br/>(coverage-risk, stability, harm checks)"]; E --> F["Regression tests and versioned artefacts<br/>Prevent unsafe drift, ensure reproducibility"];
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(controlled uncertainty, Python modules)

Deterministic logic execution
(TRI-X core inference)

XAI analysis
(SHAP / NMF / counterfactual / trace)

Safety-oriented evaluation
(coverage-risk, stability, harm checks)

Regression tests and versioned artefacts
Prevent unsafe drift, ensure reproducibility